

Vincent Wang

Robotics Systems | Vision-Language-Action Models | VLA Post-training

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EDUCATION

University of Pennsylvania

Expected May 2027

M.S. in Mechanical Engineering and Applied Mechanics - Mechatronics and Robotics

- Coursework: Advanced Dynamics, Mechatronic Systems, Engineering Mathematics, Operating Systems.

East China University of Science and Technology

June 2025

B.Eng. in Intelligent Manufacturing Engineering | GPA: 3.6/4.0 | Top 15%

EXPERIENCE

Synthoid.ai | VLA Research Intern

May 18 - Aug 21, 2026

Real-robot infrastructure, VLA training, and independent policy-distillation research

- Independently built the robot-side SDK control and VLA adapter for a 7-DoF industrial arm, adding Cartesian end-effector control, multi-rate interpolation, and automatic recovery; delivered stable human teleoperation.
- Led teleoperation and real-robot data collection with a dexterous hand, head/wrist cameras, and LeRobot; synchronized multi-modal timestamps and screened abnormal trajectories with first- through third-order action derivatives.
- Deeply contributed to early training of a proprietary VLA model through training implementation and configuration, distributed runs, checkpoint management, initial ablations, and failure diagnosis; identified a gripper-dimension output-contract mismatch.
- Independently defined and implemented Path-OPD for flow-matching VLA distillation. Under matched on-policy occupancy, Teacher-query, training, and evaluation budgets, it achieved higher closed-loop success than endpoint DAGger. Preparing an ICLR 2027 submission.

Shanghai Wuji Technology | Robotics Control Intern

July - Sept 2024

Dexterous-hand teleoperation and low-level motor control

- Re-architected an unstable impedance loop into an admittance controller with torque feedback in MATLAB/Simulink for dexterous-hand teleoperation.
- Developed Linux C++ host software for reliable CAN communication and low-level control across approximately 30 motors.

SELECTED RESEARCH & LEADERSHIP

ECUST Robocon Team | Co-founder & Control Lead

Sept 2022 - July 2024

Autonomous robot systems and interdisciplinary technical leadership

- Built the team's first STM32 motion-control system and integrated Point-LIO, YOLO (approximately 20 FPS embedded), MoveIt, and chassis control into an autonomous robot stack.
- Led a 20-member interdisciplinary team, owned system architecture and technology selection, and directed technical training for 60 new members.

MDATS - Multi-Domain Adaptive Classification for Mechanical Fault Detection

IEEE MEAE 2024

- First author; built the PyTorch research pipeline and achieved 88.27% target-domain accuracy, a 40.6-point absolute improvement over baseline.

TECHNICAL SKILLS

Languages & Systems: Python, C/C++, Rust, Linux, Git, CMake, ROS 2, CAN

Robotics & Control: Cartesian control, teleoperation, impedance/admittance control, STM32/ESP32, FreeRTOS, MoveIt

Learning & Research: PyTorch, distributed training, VLA post-training, flow matching, policy distillation, LeRobot, experimental design